Performance Comparison and Improvement for VLP With a Single LED and a Single Rotatable PD

Hanyang Shi, Caiyi Wang, and Yan Zhao

***Abstract*—We investigate two indoor visible light positioning (VLP) systems consisting of a single light-emitting diode (LED) and a single rotatable photodetector (PD) based on the received signal strength. In order to select the better one from the two VLP systems in a single LED and a single PD scenarios, we need to compare the positioning performance of two VLP systems under any noise conditions. Therefore, considering signal-dependent shot noise (SDSN), we derive the Cramér-Rao lower bound (CRLB) on the positioning error of the two VLP systems. Based on the better VLP system, we propose the adaptive-tilt-angle-adjustment (ATAA) localization algorithm by combining the least-square (LS) localization algorithm. Simulation results show that compared to LS localization algorithm, ATAA localization algorithm clearly has higher positioning accuracy.**

***Index Terms*—Visible light positioning (VLP), photodetector (PD), the Cramér-Rao lower bound (CRLB), least-square (LS).**

I. Introduction

IN RECENT years, the demand for indoor high-precision positioning technology has been continuously increasing. Due to the satellite signal attenuation caused by walls and multipath propagation, Global Positioning System (GPS) has low indoor positioning accuracy [1]. Compared to GPS and other indoor positioning technologies, visible light positioning (VLP) based on light-emitting diode (LED) is considered as a promising indoor positioning technique due to the remarkable advantages of high positioning accuracy, low implementation cost and high security [2].

In VLP systems, thermal noise and shot noise are two main sources of noise. Thermal noise results from electron excitation within the photodetector (PD), and is therefore independent of the signal and follows a Gaussian distribution. Shot noise occurs due to photon interactions with the PD, and is therefore related to signal strength. Thus, it is termed signal-dependent shot noise (SDSN). Although SDSN is Poisson-distributed, it is often modeled as Gaussian distribution due the large number of photons [3], [4], [5].

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According to the required number of lamps, VLP technology can be further divided into multilamps positioning schemes and single-lamp positioning schemes. The single-lamp positioning schemes is an indispensable part of indoor VLP research [6]. The authors in [7] describe an indoor VLP system of the single-tilted-rotatable-photodetector (STRP)-based receiver in single-LED scenario, and propose an improved gray wolf optimization (IGWO)-based localization algorithm with the assistance of the least-square (LS) method. Meanwhile, in [8], the authors design an indoor VLP system of the single-horizontal-rotatable-photodetector (SHRP)-based receiver in single-LED scenario, and propose an indoor VLP algorithm based on machine learning (ML) methods. Additionally, in [9], the authors propose a LS localization algorithm based on the SHRP model in [8], which has higher positioning accuracy and fewer rotation times compared to the positioning algorithm in [8].

The main contributions of our work can be outlined as follows: 1) In order to compare the positioning performance of two VLP systems with a single LED and a single rotatable PD in [7] and [9] under any noise conditions, we derive the Cramér-Rao lower bound (CRLB) on the positioning error of the two VLP systems considering SDSN. It is worth noting that although there are related deductions of CRLB for two VLP systems in [7] and [9], they are different from the CRLB we derived in this letter. The CRLB in [7] and [9] is derived under specific fixed values of shot noise and thermal noise, which is insufficient for comparing the positioning performance of the two systems. The CRLB we derived covers all variations of shot noise and thermal noise, enabling a comprehensive comparison of the positioning performance. This is the first letter comparing the positioning performance of the two VLP systems, and provides a good reference for us to select the better one from the two VLP systems in a single LED and a single PD scenarios. 2) By combining the LS localization algorithm, we propose the adaptive-tilt-angle-adjustment (ATAA) localization algorithm based on the better VLP system. Simulation results show that compared to LS localization algorithm in [7], ATAA localization algorithm exhibits significantly lower maximum, minimum, and average positioning errors, demonstrating a marked improvement in positioning accuracy.

II. Comparative Analysis

A. System Model

We consider two different indoor VLP systems: STRP model in [7] and SHRP model in [9]. In this letter, we

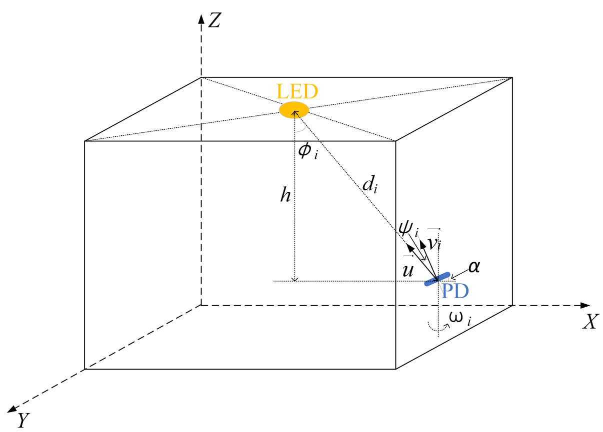

Fig. 1. STRP model.

only redescribe STRP model, and SHRP model is completely consistent with [9]. STRP model is illustrated in Fig. 1. The vertical distance from the PD to the ceiling is h, and α denotes the tilt angle of the PD. $\vec{v_i}$ is the normal vector of the plane where PD is located at ith rotation. The rotation direction of the PD is shown in Fig. 1.

In the STRP model, the tilt angle of PD is 20 degrees, and PD needs to be rotated 4 times. The first time we need to rotate the PD to the position that $\vec{v_i}$ is parallel to the X-axis, and the last three times rotate PD 90 degrees each time. In the SHRP model, the rod length is 20cm, and PD also needs to be rotated 4 times. The first time we need to rotate the PD to the position that the rod is parallel to the X-axis, and the last three times rotate PD 90 degrees each time.

We mark the three dimensional position of the LED as (X_t, Y_t, Z_t), the real position as (X, Y, Z), and the estimated position as (x, y, Z). Therefore, we have:

$$\vec{u} = \frac{1}{d}(X_t - X,\ Y_t - Y,\ Z_t - Z) \tag{1}$$

$$\begin{aligned}\vec{v_i} &= \begin{bmatrix} \cos\omega_i & \sin\omega_i & 0 \\ -\sin\omega_i & \cos\omega_i & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sin\alpha \\ 0 \\ \cos\alpha \end{bmatrix} \\ &= [\cos\omega_i \sin\alpha,\ -\sin\omega_i \sin\alpha,\ \cos\alpha] \end{aligned} \tag{2}$$

$$\omega_i = \frac{\pi(i-1)}{2} \tag{3}$$

where i=1,2,3,4 represents the number of rotations.

Considering that thermal noise and SDSN affect the optical signal s sent from the LED, the received signal can be written as [3]:

$$r = \hbar s + \sqrt{\hbar s}\, n_{\mathrm{ds}} + n \tag{4}$$

where $\hbar$ is the channel gain, n represents the signal-independent thermal noise, $n \sim N(0, \sigma_{\mathrm{n}}^2)$, and $\sqrt{\hbar s}\, n_{\mathrm{ds}}$ denotes the SDSN, $n_{\mathrm{ds}} \sim N(0, \sigma_{\mathrm{ds}}^2)$. We can define a new term $\xi^2 = \frac{\sigma_{\mathrm{ds}}^2}{\sigma_{\mathrm{n}}^2}$, and call it the shot noise scaling factor, which is used to indicate the strength of the SDSN compared with the signal-independent thermal noise. The term ξ^2, which can be determined by the receiver parameters, takes practical values ranging between 1 and 10.

B. CRLB Considering SDSN

Let λ=(X, Y). Based on (4), the probability density function (PDF) of r is:

$$f(r|\lambda) = \frac{1}{\sqrt{2\pi\sigma_{\mathrm{n}}^2(1+\xi^2\hbar s)}} \mathrm{e}^{-\frac{(r-\hbar s)^2}{2\sigma_{\mathrm{n}}^2(1+\xi^2\hbar s)}} \tag{5}$$

In the STRP model and SHRP model, the signal received by 4 different rotations are independent of each other. Therefore, we have the joint PDF $F(r_i|\lambda)$ as:

$$F(r_i|\lambda) = \prod_{i=1}^{4} \frac{1}{\sqrt{2\pi\sigma_{\mathrm{n}i}^2(1+\xi^2\hbar_i s_i)}} \mathrm{e}^{-\frac{(r_i-\hbar_i s_i)^2}{2\sigma_{\mathrm{n}i}^2(1+\xi^2\hbar_i s_i)}} \tag{6}$$

Taking the natural logarithm of (6), we obtain:

$$\begin{aligned}\ln[F(r_i|\lambda)] = &-\frac{1}{2}\sum_{i=1}^{4}\ln(2\pi\sigma_{\mathrm{n}i}^2) - \frac{1}{2}\sum_{i=1}^{4}\ln(1+\xi^2\hbar_i s_i) \\ &- \frac{1}{2\sigma_{\mathrm{n}i}^2}\sum_{i=1}^{4}\frac{(r_i-\hbar_i s_i)^2}{1+\xi^2\hbar_i s_i}\end{aligned} \tag{7}$$

The Fisher information matrix (FIM) [10] is given by:

$$F = \begin{bmatrix} -\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial X \partial X}\right\} & -\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial X \partial Y}\right\} \\ -\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial Y \partial X}\right\} & -\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial Y \partial Y}\right\} \end{bmatrix} \tag{8}$$

Note that $\mathrm{E}\{r_i\}$=$\hbar_i s_i$, and $\mathrm{E}\{(r_i - \hbar_i s_i)^2\} = \sigma_{\mathrm{n}i}^2(1+\xi^2\hbar_i s_i)$. Observing (7), we have:

$$\begin{aligned}-\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial X^2}\right\} &= \frac{\xi^2}{2}\sum_{i=1}^{4}(-\xi^2 R^2 M^2 + RJ) \\ &+ \frac{1}{2\sigma_{\mathrm{n}i}^2}\sum_{i=1}^{4}(2M^2R + 2\xi^4\sigma_{\mathrm{n}i}^2 M^2R^2 - \xi^2\sigma_{\mathrm{n}i}^2 RJ)\end{aligned} \tag{9}$$

$$\begin{aligned}-\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial X \partial Y}\right\} &= \frac{\xi^2}{2}\sum_{i=1}^{4}(-\xi^2 R^2 NM + RK) \\ &+ \frac{1}{2\sigma_{\mathrm{n}i}^2}\sum_{i=1}^{4}(2NMR + 2\xi^4\sigma_{\mathrm{n}i}^2 NMR^2 - \xi^2\sigma_{\mathrm{n}i}^2 RK)\end{aligned} \tag{10}$$

$$\begin{aligned}\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial Y \partial X}\right\} &= \frac{\xi^2}{2}\sum_{i=1}^{4}(-\xi^2 R^2 MN + RP) \\ &+ \frac{1}{2\sigma_{\mathrm{n}i}^2}\sum_{i=1}^{4}(2MNR + 2\xi^4\sigma_{\mathrm{n}i}^2 MNR^2 - \xi^2\sigma_{\mathrm{n}i}^2 RP)\end{aligned} \tag{11}$$

$$\begin{aligned}-\mathrm{E}\left\{\frac{\partial^2 \ln[\mathrm{F}(r_i|\lambda)]}{\partial Y^2}\right\} &= \frac{\xi^2}{2}\sum_{i=1}^{4}(-\xi^2 R^2 N^2 + RQ) \\ &+ \frac{1}{2\sigma_{\mathrm{n}i}^2}\sum_{i=1}^{4}(2N^2R + 2\xi^4\sigma_{\mathrm{n}i}^2 N^2R^2 - \xi^2\sigma_{\mathrm{n}i}^2 RQ)\end{aligned} \tag{12}$$

where $M = \frac{\partial(\hbar_i s_i)}{\partial X}$, $N = \frac{\partial(\hbar_i s_i)}{\partial Y}$, $J = \frac{\partial^2(\hbar_i s_i)}{\partial X^2}$, $K = \frac{\partial^2(\hbar_i s_i)}{\partial X \partial Y}$, $P = \frac{\partial^2(\hbar_i s_i)}{\partial Y \partial X}$, $Q = \frac{\partial^2(\hbar_i s_i)}{\partial Y^2}$ and $R = \frac{1}{1+\xi^2\hbar_i s_i}$. The value of $\hbar_i s_i$ is consistent with the value of I_{r} in equation (3) of [9].

In the SHRP model, $\frac{\partial(\mathrm{h}_i s_i)}{\partial X}$ and $\frac{\partial(\mathrm{h}_i s_i)}{\partial Y}$ are consistent with equation (19) and equation (20) in [9].

In the STRP model, we have:

$$\frac{\partial(\hbar_i s_i)}{\partial X} = \eta d^{-\mathrm{m}-5}(X - X_t)\Big[\cos\omega_i \sin\alpha(X_t - X) - \sin\omega_i \sin\alpha(Y_t - Y) + \cos\alpha\, h\Big] - \eta h^{\mathrm{m}} d^{-\mathrm{m}-3}\cos\omega_i \sin\alpha \tag{13}$$

$$\frac{\partial(\hbar_i s_i)}{\partial Y} = \eta d^{-\mathrm{m}-5}(Y - Y_t)\Big[\cos\omega_i \sin\alpha(X_t - X) - \sin\omega_i \sin\alpha(Y_t - Y) + \cos\alpha\, h\Big] + \eta h^{\mathrm{m}} d^{-\mathrm{m}-3}\sin\omega_i \sin\alpha \tag{14}$$

where $d = \sqrt{(X - X_t)^2 + (Y - Y_t)^2 + h^2}$, $\eta = \mathrm{R_{pd}P_{led}}\frac{(\mathrm{m}+1)\mathrm{A_{pd}}}{2\pi}\mathrm{T}(\psi)\mathrm{g}(\psi)h^{\mathrm{m}}(-\mathrm{m} - 3)$. The definitions of other parameters are consistent with [9].

$\frac{\partial^2(\hbar_i s_i)}{\partial X^2}$, $\frac{\partial^2(\hbar_i s_i)}{\partial X \partial Y}$, $\frac{\partial^2(\hbar_i s_i)}{\partial Y \partial X}$ and $\frac{\partial^2(\hbar_i s_i)}{\partial Y^2}$ can be calculated based on $\frac{\partial(\hbar_i s_i)}{\partial X}$ and $\frac{\partial(\hbar_i s_i)}{\partial Y}$.

Since localization can be considered as an unbiased estimation problem, we can derive the lower bound of the expected estimation error $\mathrm{E}[(X - x)^2 + (Y - y)^2]$:

$$\mathrm{E}[(X - x)^2 + (Y - y)^2] \geq \mathrm{trace}(F^{-1}) = \mathrm{CRLB} \tag{15}$$

III. ATAA Localization Algorithm

In the second paragraph of Section IV, we have demonstrated that STRP model has better localization performance than the SHRP model. In this section, we propose the ATAA localization algorithm based on the STRP model.

In Fig.7 of [7], we can know that as the tilt angle α becomes larger, the value of CRLB will become smaller, which means the positioning accuracy will be higher. This is because as the tilt angle α increases, there is a greater difference in the power values of the optical signals collected through 4 rotations. However, due to the limitation of PD maximum field of view angle, the tilt angle α cannot be infinitely increased, which means that there is an optimal tilt angle α_o at each positioning point. Therefore, in our improved STRP model, the tilt angle α can adaptively change.

In the ATAA localization algorithm, first, we give the tilt angle α an initial value of 20 degrees. Then, we use the LS localization algorithm in [7] for positioning. Since [7] has already described the LS localization algorithm in detail, we do not make any derivation here, but directly provide the relevant equations.

$$A = \begin{bmatrix} I_2 \sin\alpha & I_1 \sin\alpha \\ (I_3 + I_1)\sin\alpha & 0 \\ I_4 \sin\alpha & -I_1 \sin\alpha \end{bmatrix} \tag{16}$$

$$B = \begin{bmatrix} (I_2 - I_1)\cos\alpha\, h + I_2 \sin\alpha\, X_t + I_1 \sin\alpha\, Y_t \\ (I_3 - I_1)\cos\alpha\, h + (I_1 + I_3)\sin\alpha\, X_t \\ (I_4 - I_1)\cos\alpha\, h + I_4 \sin\alpha\, X_t - I_1 \sin\alpha\, Y_t \end{bmatrix} \tag{17}$$

$$B = A\begin{bmatrix} x \\ y \end{bmatrix} \tag{18}$$

where I_i represents the actual current value received at the ith rotation. The definition of I_i is consistent with equation (5) in [9].

Therefore, the estimated position of the receiver can be obtained by (19):

$$\begin{bmatrix} x \\ y \end{bmatrix} = \left(A^T A\right)^{-1} A^T B \tag{19}$$

Next, we need to find the optimal tilt angle α_o at the real position.

Based on (1), (2), we have:

$$\cos\psi_i = \vec{u}\vec{v_i} = \frac{1}{d}\cos\omega_i \sin\alpha(X_t - X) - \frac{1}{d}\sin\omega_i \sin\alpha(Y_t - Y) + \frac{1}{d}\cos\alpha h \tag{20}$$

Assuming Ψ represents the maximum field of view angle of PD, α_i represents the maximum tilt angle that can be achieved at the ith rotation, $\alpha_i \in \left(0, \frac{\pi}{2}\right)$, we have:

$$\left[\frac{1}{d}\cos\omega_i(X_t - X) - \frac{1}{d}\sin\omega_i(Y_t - Y)\right]\sin\alpha_i + \frac{h}{d}\cos\alpha_i = \cos\Psi \tag{21}$$

let $C = \left[\frac{1}{d}\cos\omega_i(X_t - X) - \frac{1}{d}\sin\omega_i(Y_t - Y)\right]$, $G = \frac{h}{d}$ and $L = \cos\Psi$.

Based on (21), we have:

$$\alpha_i = \arcsin\frac{L}{\sqrt{C^2 + G^2}} - \arctan\left(\frac{G}{C}\right) \tag{22}$$

Assuming α_e is an estimated value close to α_o. By substituting (x, y) into equation (21), we have:

$$\alpha_e = \min\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} \tag{23}$$

The value of α_e may be larger or smaller than α_o. We adjust the tilt angle from 20 degrees to α_e and rotate the PD in sequence. If at least one rotation fails to receive the signal, it indicates that α_e is larger than α_o. Otherwise, α_e is smaller than α_o. If α_e is larger than α_o, we decrease the tilt angle by one degree each time until the signal can be received after 4 rotations. At this moment, it is the optimal tilt angle α_o. If α_e is smaller than α_o, we increase the tilt angle by one degree each time until at least one rotation cannot receive the signal. At this moment, reducing the tilt angle by one degree is the optimal tilt angle α_o. After obtaining α_o, we use the LS localization algorithm for positioning again, which can achieve higher positioning accuracy.

IV. Simulation Results

Computer simulation is conducted on the MATLAB platform. Our point selection rule in the plane is as follows: the range of X and Y is from 0.2 to 4.8, with a step size of 0.2, and there are a total of 576 uniformly distributed points. Simulation parameters are consistent with [9].

Fig.2 discusses the variation of Ratio-CRLB(the CRLB of SHRP model divided by the CRLB of STRP model) when the value of ξ^2 is 1, 2, ..., 10 and h = 3m. From the figure, we can find that when the LED power is the same, the CRLB of the SHRP model is approximately 4.3 times that of the CRLB of the STRP model. This indicates that, regardless of variations in thermal noise and shot noise, the positioning performance of the STRP model is significantly better than that of the SHRP model. Both models require 4 rotations, and the greater the difference between the power values of the optical signals obtained through 4 rotations, the better the positioning effect. In the SHRP model, the rod length leads to different optical channel gain obtained from 4 rotations. In the STRP model, the tilt angle leads to different optical channel gain obtained

Algorithm 1 The Proposed ATAA Localization Algorithm

Set the initial value of α to 20, and rotate PD 4 times to get I_1, I_2, I_3, I_4;
Substitute I_1, I_2, I_3, I_4 into equation(19) to get (x, y);
Substitute (x, y) into equation(23) to get α_e;
If $\alpha_e > \alpha_o$
  Decrease α_e by one degree each time until reaching α_o;
Else
  Increase α_e by one degree each time until reaching α_o;
End
Rotate PD 4 times again to obtain new I_1, I_2, I_3, I_4, substitute α_o, I_1, I_2, I_3, I_4 into equation(19) to get the optimal (x, y);

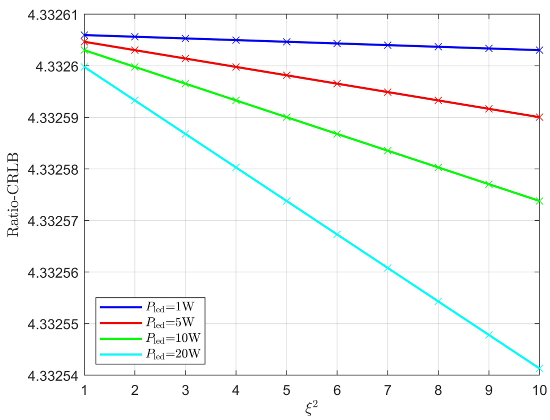

Fig. 2. The variation of Ratio-CRLB when the value of ξ^2 is 1, 2, ..., 10.

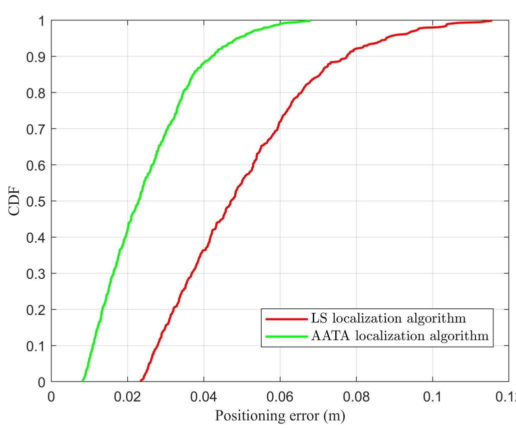

Fig. 3. Comparison of CDF of the positioning errors under two algorithms.

from 4 rotations. The impact of changing the tilt angle is much greater than that of changing the rod length, which is why STRP model is much better than the SHRP model.

We define the error function as: $d_{\text{error}} = \sqrt{(X - x)^2 + (Y - y)^2}$. Since ATAA localization algorithm is an improvement based on LS localization algorithm, we only compared it with LS localization algorithm. Fig.3 compares the CDF of the positioning errors under two localization algorithms when h = 3m and P_{led} = 5W. In the LS localization algorithm, the maximum positioning error is 0.1154m, the minimum positioning error is 0.0234m, and the average positioning error is 0.0503m. In the ATAA localization algorithm, the maximum positioning error is 0.0680m, the minimum positioning error is 0.0082m, and the average positioning error is 0.0250m, all of which are much smaller than the positioning errors of the LS localization algorithm. This indicates that the positioning accuracy of ATAA localization algorithm is much higher than that of LS localization algorithm in [7].

V. Conclusion

In this letter, we compare the positioning performance of two indoor VLP systems consisting of a single LED and a single rotatable PD based on the received signal strength in [7] and [9]. We compare two VLP systems under any noise conditions by deriving the CRLB on the positioning error of the two VLP systems considering SDSN. The simulation results show that the STRP model has better positioning performance. By combining the LS localization algorithm, we propose the ATAA localization algorithm based on the STRP model. The simulation results show that ATAA localization algorithm clearly has higher positioning accuracy compared to LS localization algorithm in [7]. Future research includes studying the impact of environmental noises on the system and explore how to implement the adaptive adjustment STRP receiver in hardware.

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